



第五章 留数

作业

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1. $z = 0$ 是函数 $f(z) = \frac{1}{\cos\left(\frac{1}{z}\right)}$ 的孤立奇点吗？为什么？

解： $\cos\left(\frac{1}{z}\right) = 0 \Rightarrow \frac{1}{z} = k\pi + \frac{\pi}{2}, k = 0, \pm 1, \pm 2, \dots$

$$\Rightarrow z = \frac{1}{k\pi + \frac{\pi}{2}} \text{ 为奇点}$$

当 $k \rightarrow \infty$ 时, $z \rightarrow 0$, 即在 $z = 0$ 的无论怎样小的去心邻域内,

总有 $f(z)$ 的其它奇点 $z = \frac{1}{k\pi + \frac{\pi}{2}} (k \rightarrow \infty)$ 存在,

$\therefore z = 0$ 不是孤立奇点.

3.下列函数有些什么奇点？如果是极点，指出其级数。

$$(5) \frac{\sin z}{z^3}; \quad (8) \frac{1}{e^z - 1};$$

解：

$$(5) \frac{\sin z}{z^3} = \frac{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots}{z^3} = z^{-2} - \frac{1}{3!} + \frac{z^2}{5!} - \cdots$$

$\therefore z = 0$ 是二级极点

$$(8) e^z - 1 = 0 \Rightarrow z = \operatorname{Ln} 1 = 2k\pi i, k = 0, \pm 1, \pm 2 \cdots$$

$z = 2k\pi i$ 时,

$$e^z - 1 = 0,$$

$$(e^z - 1)' = e^z = 1 \neq 0,$$

$\therefore z = 2k\pi i$ 是 $e^z - 1$ 的一级零点

$\therefore z = 2k\pi i (k = 0, \pm 1, \pm 2 \cdots)$ 是 $\frac{1}{e^z - 1}$ 的一级极点.

7.求下列各函数在有限奇点处的留数。

$$(2) \frac{1-e^z}{z^3}; \quad (4) \cos \frac{1}{1-z}; \quad (6) \frac{z}{\cos z};$$

解: (2) $\frac{1-e^z}{z^3} \Rightarrow z^3 = 0 \Rightarrow z = 0$ 是极点 (实际是2级极点, 这里取高1级)

$$\operatorname{Res}[f(z), 0] = \frac{1}{2!} \left(z^3 \cdot \frac{1-e^z}{z^3} \right)'' \Big|_{z=0} = \frac{1}{2} \cdot (-e^z) \Big|_{z=0} = -\frac{1}{2}$$

(4) $\cos \frac{1}{1-z}$, 在 $0 < |z-1| < +\infty$ 内展开成洛朗级数

$$= \cos \frac{1}{z-1} = 1 - \frac{\left(\frac{1}{z-1}\right)^2}{2!} + \frac{\left(\frac{1}{z-1}\right)^4}{4!} - \cdots = 1 - \frac{1}{2!(z-1)^2} + \frac{1}{4!(z-1)^4} - \cdots$$

有无穷多负幂项 $\Rightarrow z = 1$ 是本性奇点且 $c_{-1} = 0$

$$\therefore \operatorname{Res}[f(z), 1] = 0.$$

$$(6) \frac{z}{\cos z}, \quad \cos z = 0 \Rightarrow z = k\pi + \frac{\pi}{2}, k = 0, \pm 1, \pm 2, \dots$$

$$\left(\frac{\cos z}{z} \right)' \Big|_{z=k\pi+\frac{\pi}{2}} = \frac{-\sin z \cdot z - \cos z}{z^2} \Big|_{z=k\pi+\frac{\pi}{2}} \neq 0$$

$\therefore z = k\pi + \frac{\pi}{2}$ 是 $\frac{z}{\cos z}$ 的一级极点

$$\operatorname{Res} \left[f(z), k\pi + \frac{\pi}{2} \right] = \frac{P(z_0)}{Q'(z_0)} = \frac{z}{(-\sin z)} \Big|_{z=k\pi+\frac{\pi}{2}}$$

$$= \begin{cases} -(k\pi + \frac{\pi}{2}), k = 0, \pm 2, \pm 4, \dots \\ k\pi + \frac{\pi}{2}, k = \pm 1, \pm 3, \pm 5, \dots \end{cases}$$

$$= (-1)^{k+1} (k\pi + \frac{\pi}{2}), \quad k = 0, \pm 1, \pm 2, \dots$$

8.利用留数计算下列积分（提示：圆周取正向）。

$$(3) \oint_{|z|=2} \frac{e^z}{z(z-1)^2} dz;$$

解： 可以判断 $z=0$ 是一级极点, $z=1$ 是二级极点

$$\operatorname{Res}[f(z), 0] = \lim_{z \rightarrow 0} \frac{e^z}{(z-1)^2} = 1$$

$$\operatorname{Res}[f(z), 1] = \left(\frac{e^z}{z} \right)' \bigg|_{z=1} = \left(\frac{e^z \cdot z - e^z}{z^2} \right) \bigg|_{z=1} = e - e = 0$$

$$\therefore \text{原式} = 2\pi i [\operatorname{Res}[f(z), 0] + \operatorname{Res}[f(z), 1]] = 2\pi i$$

9.判定 $z = \infty$ 是下列各函数的什么奇点？并求出在 ∞ 的留数。

$$(3) \frac{z}{2+z^2}$$

解：

$$\lim_{z \rightarrow \infty} \frac{z}{2+z^2} = \lim_{z \rightarrow \infty} \frac{1}{2z} = 0 \Rightarrow z = \infty \text{ 是可去奇点}$$

$$\operatorname{Res}[f(z), \infty] = -\operatorname{Res}\left[f\left(\frac{1}{z}\right) \cdot \frac{1}{z^2}, 0\right]$$

$$= -\operatorname{Res}\left[\frac{\frac{1}{z}}{2+\frac{1}{z^2}} \cdot \frac{1}{z^2}, 0\right] = -\operatorname{Res}\left[\frac{1}{z(2z^2+1)}, 0\right]$$

$$= -\lim_{z \rightarrow 0} \frac{1}{2z^2+1} = -1$$

10. 计算下列各积分, C 为正向圆周。

$$(1) \oint_C \frac{z^{15}}{(z^2 - 1)^2 (z^4 + 2)} dz, C : |z| = 3;$$

解: $f(z) = \frac{z^{15}}{(z^2 - 1)^2 (z^4 + 2)}$ 在 $|z| = 3$ 外只有奇点 $z = \infty$

在 $3 < |z| < +\infty$ 域内,

$$\begin{aligned} f(z) &= z^7 \left(\frac{1}{1 - z^{-2}} \right)^2 \frac{1}{(1 + 2z^{-4})} \\ &= z^7 (1 + z^{-2} + z^{-4} + \dots)^2 (1 - 2z^{-4} + 4z^{-8} - \dots) \\ &= z^7 (1 + 2z^{-2} + 3z^{-4} + 4z^{-6} + 5z^{-8} + \dots) (1 - 2z^{-4} + 4z^{-8} - \dots) \\ &= z^7 (1 + 2z^{-2} + z^{-4} + 3z^{-8} + \dots) \\ &= z^7 + 2z^5 + z^3 + 3z^{-1} + \dots \end{aligned}$$

$$\therefore \text{原式} = -2\pi i \operatorname{Res}[f(z), \infty] = -2\pi i(-3) = 6\pi i$$

11.计算下列积分:

$$(1) \int_0^{2\pi} \frac{1}{5+3\cos\theta} d\theta$$

$$= \oint_{|z|=1} \frac{1}{5 + \frac{3(z^2+1)}{2z}} \frac{dz}{iz}$$

$$= \oint_{|z|=1} \frac{-2i}{10z + 3z^2 + 3} dz = \oint_{|z|=1} \frac{-2i}{(3z+1)(z+3)} dz$$

$$= 2\pi i \operatorname{Res} \left[\frac{-2i}{(3z+1)(z+3)}, -\frac{1}{3} \right]$$

$$= 2\pi i \lim_{z \rightarrow -\frac{1}{3}} \frac{-2i}{z+3} = \frac{1}{2} \pi$$

12.计算下列积分:

$$(1) \int_{-\infty}^{+\infty} \frac{\cos x}{x^2 + 4x + 5} dx; \quad (2) \int_0^{+\infty} \frac{x^2}{1+x^4} dx;$$

(1)解: 令 $f(z) = \frac{e^{iz}}{z^2 + 4z + 5} = \frac{e^{iz}}{(z+2+i)(z+2-i)}$, 上半平面一级极点 $z = -2+i$

$$\operatorname{Res}[f(z), -2+i] = \lim_{z \rightarrow -2+i} \frac{e^{iz}}{z+2+i} = \frac{e^{-1-2i}}{2i} = -\frac{e^{-1-2i}}{2} i = -\frac{\sin 2}{2e} - \frac{\cos 2}{2e} i$$

$$\therefore \text{原式} = \operatorname{Re}(2\pi i \operatorname{Res}[f(z), -2+i])$$

$$= \operatorname{Re} \left[2\pi i \cdot \left(-\frac{\sin 2}{2e} - \frac{\cos 2}{2e} i \right) \right]$$

$$= \operatorname{Re} \left[\frac{\pi}{e} \cos 2 - \left(\frac{\pi}{e} \sin 2 \right) i \right]$$

$$= \frac{\pi}{e} \cos 2$$

$$(2) \int_0^{+\infty} \frac{x^2}{1+x^4} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^2}{1+x^4} dx$$

$$\text{令 } f(z) = \frac{z^2}{1+z^4},$$

$$z^4 = -1 \Rightarrow z = \cos \frac{2k\pi - \pi}{4} + i \sin \frac{2k\pi - \pi}{4}, k = 0, 1, 2, 3$$

$$\therefore \text{上半平面一级极点 } z_1 = \frac{\sqrt{2}}{2}(1+i), \quad z_2 = \frac{\sqrt{2}}{2}(-1+i)$$

$$\text{Res}[f(z), z_1] = \frac{z^2}{(1+z^4)'} \Big|_{z=\frac{\sqrt{2}}{2}(1+i)} = \frac{\frac{1}{2}(2i)}{4 \left[\frac{\sqrt{2}}{2}(1+i) \right]^3} = \frac{\sqrt{2}}{8}(1-i)$$

$$\text{Res}[f(z), z_2] = \frac{z^2}{(1+z^4)'} \Big|_{z=\frac{\sqrt{2}}{2}(-1+i)} = -\frac{\sqrt{2}}{8}(1+i)$$

$$\therefore \text{原式} = \frac{1}{2} \cdot 2\pi i \cdot \left[\frac{\sqrt{2}}{8}(1-i) - \frac{\sqrt{2}}{8}(1+i) \right] = \pi i \cdot \frac{-\sqrt{2}}{4} i = \frac{\sqrt{2}}{4} \pi$$